

Homework week 7, core econometrics, fall 2025

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1. Partial identification:

Suppose the following assumptions hold:

- (a) $Y = D \cdot Y^1 + (1 - D) \cdot Y^0$, where $D \in \{0, 1\}$ and $Y \in [0, 1]$,
- (b) $D = Z \cdot D^1 + (1 - Z) \cdot D^0$, where $Z \in \{0, 1\}$,
- (c) $D^1 \geq D^0$,
- (d) $Z \perp (Y^0, Y^1, D^0, D^1)$.

Provide explicit bounds on the CDF of potential outcomes,

$$P(Y^d \leq y),$$

for $d = 0, 1$. Discuss these bounds in terms of “never takers”, “always takers”, and “compliers”.

2. Inverse probability weighting:

Suppose the following assumptions hold:

- (a) $D \in \{0, 1\}$
- (b) $Y = D \cdot Y^1 + (1 - D) \cdot Y^0$
- (c) $(Y^1, Y^0) \perp D | X$
- (d) $p(X) < 1$ for almost all X , where $p(X) = P(D = 1 | X)$

Find a weight function $w(D, X)$, such that

$$E[w(D, X) \cdot \mathbf{1}(Y \leq y)] = P(Y^0 \leq y | D = 1).$$

3. Delta method:

Consider i.i.d. draws of a random variable X_i with mean μ and variance σ^2 .

- (a) What is the asymptotic distribution of $\sqrt{n} \cdot (\bar{X}_n - \mu)$, where $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$?

- (b) What is the asymptotic distribution of $\sqrt{n} \cdot (|\bar{X}_n| - |\mu|)$, for $\mu \neq 0$?
(Hint: Use the delta method.)
- (c) What happens when $\mu = 0$?
What happens when μ is close to 0?
(Hint: Explicitly consider the remainder of the first-order approximation used for the delta method.)
- (d) Discuss the bias and deviation from normality of the distribution of $\bar{X}_n$, assuming that $X_i \sim N(\mu, \sigma^2)$, so that the approximation in item (a) is exact.